

TOPIC 20/60

What is Pre-training?

The phase where AI learns
the structure of human thought.

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THE GOAL

Pre-training is the "School" phase. It transforms raw internet data into a model that understands language, logic, and common-sense reasoning.

The Core Objective: Self-Supervised Learning

The model creates its own labels by predicting missing data.

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HOW IT WORKS



Predict the Next Token

This is the standard pre-training method. You take trillions of words and mask the final one.

"The apple fell from the [?]"

If it guesses "tree", it gets a reward. If it guesses "car", it gets penalized.

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THE VOLUME

Global Consumption

Pre-training is not about reading a few books. It requires "The Internet."

The Data Source:

Web pages (Common Crawl), Wikipedia, GitHub (Code), Books (Project Gutenberg), and Academic Journals.

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COMPUTE

Brute Force

This phase is the most expensive part of AI.

Thousands of H100 GPUs run for months on end. They calculate the error in billions of predictions and adjust every single parameter to reduce that error.

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EMERGENCE

Emergent Skills

Something strange happens at massive scale.

The scaling law:

As you increase data, compute, and model size, the AI doesn't just get better at prediction—it suddenly starts **reasoning, coding, and translating.**

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WHY PRE-TRAINING IS NOT "FINE-TUNING"

PRE-TRAINING

Goal: Learn world knowledge.

Result: A generic "Base Model". It finishes sentences but doesn't act like an assistant.

FINE-TUNING

Goal: Follow instructions.

Result: The "Instruct" model you actually chat with.

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THE RESULT

The Foundation

After pre-training, the model is like a human who has read every book in a library but never actually spoken to another person.

It has high "raw intelligence" but lacks the social alignment to be helpful in a chat interface. That comes next.

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THE REQUIREMENTS

Massive Compute

Millions of dollars in electricity and hardware.

Quality Data

De-duplicated, filtered, and cleaned text. High quality data = High quality model.

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Pre-training

CONCEPT UNLOCKED



Objective: Next-Token Prediction



Data: Trillions of words



Result: Foundation Model

CONNECT WITH ME



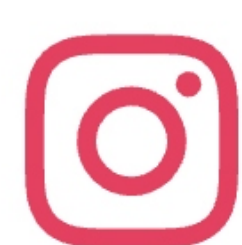
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